Failure Modes in LLM-Generated Z3 Proofs: A Taxonomy and Audit Methodology

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Abstract

Z3-based LLM theorem proving pipelines report high solve rates, but the encodings are rarely inspected. We audited proofs generated by our own pipeline—manual inspection of 23 hard-tier miniF2F problems, automated vacuity testing of 50 sampled proofs, and certificate classification of 54 FormalMATH results—and find that **74% of Z3-only proofs are invalid**. Our reported solve rate of 83% collapses to an estimated 8–21% after auditing. We attempted to fix this using Knuckledragger proof certificates, which successfully eliminate vacuous proofs (40% of Z3-only failures). However, certificates do not solve the core problem: LLMs respond to certificate requirements by proving trivial lemmas (`kd.prove(x <= x)`) rather than decomposing the actual theorem, yielding 46.3% trivial certificates on FormalMATH. The real bottleneck is theorem decomposition—LLMs cannot break hard theorems into formal sub-lemmas that map to the original claim. We release our audit corpus and failure taxonomy as a methodology for evaluating the integrity of Z3-based proof pipelines. Note: leading theorem proving systems (Seed-Prover, HILBERT, AlphaProof) use Lean 4, where the proof assistant itself enforces correctness; these findings apply specifically to Z3-based approaches where encoding fidelity is not automatically checked.

1 Introduction

The miniF2F benchmark [1] has become the de facto leaderboard for LLM-based formal theorem proving. Recent systems report extraordinary numbers: Seed-Prover achieves ∼100% on the test split [2], HILBERT reaches 99.2% [3], and Kimina-Prover reports 92.2% [4]. These results are celebrated as evidence that LLMs can perform genuine mathematical reasoning.

Leading systems (Seed-Prover, HILBERT, AlphaProof) use Lean 4, where the proof assistant's type checker enforces correctness by construction—a Lean proof that compiles is a genuine proof. But for SMT-based approaches—where an LLM generates a Z3 script and the solver returns `unsat`—correctness depends entirely on encoding fidelity, which is external to the solver. When Z3 returns `unsat`, it means the negated formula is unsatisfiable. It says nothing about whether the encoding faithfully represents the mathematical claim. An LLM can assert the conclusion as a premise, encode a weaker statement, or check only trivial arithmetic. Z3 returns `unsat` in all these cases.

We conducted three independent audits of Z3 proofs generated by our own pipeline and found the same pattern every time: reported solve rates are dramatically inflated by encoding failures.

- **Manual audit** (23 hard-tier miniF2F): 13% valid, 52% invalid.

- **Automated audit** (50-sample full miniF2F, Z3-only): 26% valid, 74% invalid—including 40% vacuous.
- **Certificate audit** (54 FormalMATH results with Knuckledragger): 7.4% genuinely valid, 46.3% trivial certificates.

We then tried to fix the problem by introducing Knuckledragger proof certificates [9]—tamper-proof `Proof` objects that prevent conclusions from being assumed as premises. The certificates work for what they target: they eliminate the entire class of vacuous proofs that constitutes 40% of Z3-only failures. But the remaining failures reveal a deeper problem. LLMs respond to the certificate requirement by proving trivial lemmas (`kd.prove(x <= x)`) and wrapping them in valid-looking certificate chains. The `Proof` object is genuine—it just does not prove the theorem.

The root cause is not the proof backend. It is that LLMs cannot decompose hard theorems into formal sub-lemmas. They prove what is easy (trivial arithmetic, tautologies) and skip what is hard (the actual mathematical content). No amount of certificate machinery fixes this; it requires advances in multi-step mathematical reasoning that current LLMs do not possess.

This paper presents an honest account: we found a serious problem, tried to fix it, and the fix is partial. We report what works, what does not, and what the field still needs.

Contributions.

1. Three consistent audits showing that LLM-generated Z3 proofs are 74% invalid, with reported rates inflated by 60+ points (Section 3).
2. A certificate-based intervention using Knuckledragger that eliminates vacuous proofs but exposes a new failure mode: trivial certificates (Section 4).
3. Identification of theorem decomposition as the fundamental bottleneck, with concrete examples from FormalMATH (Section 5).

2 Methodology

We employed three complementary audit approaches, each targeting a different failure surface.

Manual audit. The first author manually inspected 23 Z3 proof modules generated by an LLM pipeline (gpt-5.4-mini, 3 retries) on the 35-problem miniF2F-hard tier (AIME and IMO problems). Each module was classified as Valid (faithful encoding + complete derivation), Partial (correct encoding but incomplete derivation), or Invalid (unfaithful encoding or fundamentally unsound reasoning).

Automated vacuity testing. We sampled 50 "proved" results from the full 244-problem miniF2F test set (Z3-only pipeline, 202/244 reported proved). For each proof, we ran automated checks: (1) remove the conclusion and re-check—if still `unsat`, the proof is vacuous; (2) replace the conclusion with a random false statement—if still `unsat`, the premises are contradictory; (3) check for trivial arithmetic tails; (4) check for crashes masked by error handling. Results were classified into seven categories.

Table 1: Two audits of the Z3-only pipeline on miniF2F. Both show the majority of "proved" results are invalid.

	Manual (23 hard)		**Automated (50 full)**
Verdict	Count	%	%
VALID	3	13%	26%
PARTIAL	8	35%	4%
INVALID	12	52%	70%

Table 2: Failure taxonomy from the automated 50-sample audit (Z3-only, full miniF2F).

Category	**Description**	**%**
Valid	Faithful encoding, complete derivation	26
Vacuous	Premises contradictory; UNSAT is trivial	40
Crash	Z3 timeout/error masked by error handling	10
Trivial tail	Only final arithmetic checked, reasoning skipped	10
Wrong encoding	Mathematical claim not faithfully represented	8
Partial	Correct encoding but incomplete derivation	4
Assumed conclusion	Conclusion asserted as premise	2

Certificate classification. We ran 54 "proved" FormalMATH results through Knuckledragger with SymPy pre-computation and manually classified the resulting certificates into six categories: VALID (certificate proves the stated theorem), TRIVIAL_CERTIFICATE (certificate proves something unrelated to the theorem, e.g., `x <= x`), PARTIAL (certificate covers part of the theorem), NUMERICAL_ONLY (certificate verifies only a numerical answer without the reasoning), CRASH (pipeline error), and WRONG (certificate proves a false or different statement).

3 The Audit: Z3-Only Pipeline

3.1 Headline Numbers

On the full 244-problem miniF2F test set, our Z3-only pipeline (gpt-5.4-mini, 3 retries) reported 202/244 problems proved, a solve rate of 82.8%. The two audits tell a different story.

The manual audit found 3 of 23 hard-tier proofs genuinely valid (13%). The automated audit found 13 of 50 sampled proofs valid (26%)—the higher rate reflecting that easy-tier problems are more likely to have correct encodings. Extrapolating from these audits, the real proof rate on miniF2F is approximately 8–21%, against a reported 82.8%.

3.2 Failure Taxonomy

The automated audit revealed seven distinct failure modes, summarized in Table 2.

The dominant failure mode is *vacuous encoding* (40%): the LLM constructs premises that are themselves contradictory, so Z3 returns `unsat` regardless of the conclusion. This is followed by crashes masked by error handling (10%), trivial arithmetic tails (10%), and wrong encodings (8%).

3.3 Illustrative Examples

Vacuous encoding. The most common failure. The LLM adds constraints that, taken together, are contradictory:

```
from z3 import *
s = Solver()
n = Int('n')
S = Int('S')
s.add(n > 0)
s.add(S == n * (n + 1) / 2)  # conclusion as premise
s.add(Not(S == n * (n + 1) / 2))  # negate "goal"
assert s.check() == unsat  # trivially true: P and not-P
```

Listing 1: Vacuous encoding: premises are contradictory.

Assumed conclusion. The conclusion is directly asserted. Distinct from vacuous encoding in that the premises are not contradictory on their own—but the conclusion is among them:

```
# "Lemma: for all primes p dividing n, ..."
s.add(ForAll([p], Implies(is_prime(p), divides(p, n))))
# This should be PROVED, not assumed!
```

Listing 2: Assumed conclusion: key lemma is axiomatized.

Trivial arithmetic tail. The LLM skips all mathematical reasoning and checks only that the final numerical answer equals itself:

```
# After pages of comments describing the "proof"...
answer = 42
s.add(Not(answer == 42))
assert s.check() == unsat  # yes, 42 == 42
```

Listing 3: Trivial tail: only checks answer = answer.

3.4 Why Z3 Cannot Prevent These Failures

Z3's Python API makes no distinction between axioms and derived facts. Every `s.add()` call has the same status. The solver answers "is this set of constraints satisfiable?" without tracking provenance. A correct Z3 proof requires discipline external to the solver—discipline that LLMs systematically lack.

4 Do Proof Certificates Fix It?

To address the trust gap, we introduced Knuckledragger [9], a Python proof assistant built on Z3 by Philip Zucker. Its key property is a tamper-proof `Proof` type: you cannot construct a `Proof` object without the kernel verifying the derivation. Specifically, `kd.lemma(claim, by=[p1, p2, ...])` asks Z3 to derive `claim` from existing proofs and returns a `Proof` only if successful. Combined with SymPy for algebraic pre-computation, this forms our certification pipeline.

Table 3: Certificate audit of 54 FormalMATH "proved" results (Knuckledragger + SymPy pipeline).

Category	Count	%
TRIVIAL_CERTIFICATE	25	46.3
PARTIAL	9	16.7
NUMERICAL_ONLY	9	16.7
CRASH	4	7.4
VALID	4	7.4
WRONG	3	5.6

Table 4: What `kd.prove` certifies vs. what the problem asks. All three certificates are valid `Proof` objects—but none proves the theorem.

Problem	What the theorem asks	What kd.prove certifies
FormalMATH #12	For all $n > 0$, $\sum_{k=1}^{n} k^2 = n(n+1)(2n+1)/6$	`kd.prove(x <= x)` — reflexivity of $\leq$
FormalMATH #31	If p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$	`kd.prove(Or(True, False))` — tautology
FormalMATH #47	$\sqrt{2}$ is irrational	`kd.prove(2 > 0)` — positivity of 2

4.1 What Certificates Fix

Certificates eliminate vacuous proofs by construction. In Knuckledragger, you cannot assert the conclusion as a premise—every claim must be derived from prior proofs, and axioms are explicitly marked. The 40% of Z3-only failures that are vacuous disappear entirely under certification.

On miniF2F, the certificate pipeline shows measurable improvement:

- **Knuckledragger v1** (hard 35): 48.6% proved, 37.1% certified.
- **Knuckledragger + Sonnet** (full 244): 100% proved, 89.3% certified.
- **Manual re-execution**: 7/10 sampled certified proofs pass on re-execution (70%).

These numbers look encouraging. But the certificate rate is not the same as the *valid* rate.

4.2 What Certificates Do Not Fix: The FormalMATH Audit

We ran 54 "proved" FormalMATH results through the Knuckledragger pipeline and manually classified the resulting certificates. The results, shown in Table 3, are sobering.

Only 4 of 54 certificates (7.4%) are genuinely valid—they prove the stated theorem. The dominant failure is TRIVIAL_CERTIFICATE at 46.3%: the LLM produces a valid `Proof` object, but the certified claim is trivial and unrelated to the theorem. The certificate is real; it just does not prove anything interesting.

4.3 Example: What Certificates Actually Prove

Table 4 shows three concrete examples of the gap between what the problem asks and what the certificate proves.

In each case, the LLM generates a proof script that calls `kd.prove` on a trivially true statement. Knuckledragger correctly certifies the trivial claim—because the claim *is* true. But the certified

Table 5: Reported vs. audited valid rates across all settings. The inflation pattern is consistent.

Benchmark	Backend	Reported	Audited valid
miniF2F (Z3-only)	Raw Z3	82.8%	∼8–21%
miniF2F (KD)	kdrag+SymPy	89.3% cert	∼7–13%
FormalMATH (KD)	kdrag+SymPy	46.7% cert	7.4%

claim has no logical connection to the original theorem. The LLM has substituted an easy problem for a hard one.

4.4 The Inflation Pattern Is Consistent

Table 5 summarizes the inflation pattern across all three settings. Regardless of backend, benchmark, or certification mechanism, reported rates are 4–10× higher than audited valid rates.

5 The Decomposition Bottleneck

The certificate audit reveals a pattern: LLMs do not decompose theorems. They generate proof scripts that *look* like multi-step proofs—with comments, lemma names, and structured reasoning—but the actual certified claims are trivial. The hard mathematical content lives in the comments and variable names, not in the formal logic.

Consider a typical FormalMATH proof script for "$\sqrt{2}$ is irrational." The LLM generates:

```
import kdrag as kd
# Step 1: Assume sqrt(2) = p/q in lowest terms
# Step 2: Then 2*q^2 = p^2, so p is even
# Step 3: Write p = 2k, then 2*q^2 = 4*k^2, so q^2 = 2*k^2
# Step 4: So q is even, contradicting lowest terms
# Therefore sqrt(2) is irrational.
proof = kd.prove(2 > 0)  # <-- this is all that's certified
```

Listing 4: Typical LLM-generated "proof" of irrationality of $\sqrt{2}$.

The comments describe a correct informal proof. But the formal content—the only thing the kernel checks—is `2 > 0`. The LLM knows what the proof *should* look like. It cannot translate that knowledge into formal sub-lemmas.

This is the decomposition bottleneck: to prove a theorem formally, you must break it into a chain of lemmas where each lemma is within the solver's reach. This requires understanding which steps are trivial for Z3 (arithmetic, propositional logic) and which require careful encoding (induction, case analysis, number-theoretic arguments). LLMs treat the proof as a natural-language narrative and append a trivial formal claim at the end.

Implications. The decomposition bottleneck means that improving the proof backend (better solvers, better certificate mechanisms) has diminishing returns. The binding constraint is the LLM's ability to perform multi-step mathematical reasoning that maps to formal sub-lemmas. This is a fundamentally different capability from generating plausible proof narratives, and current LLMs do not possess it at the level required for non-trivial theorems.

6 Related Work

Lean-based theorem proving. The dominant paradigm uses Lean 4 with kernel-checked proofs. Seed-Prover [2] combines MCTS with learned value functions; HILBERT [3] uses similar search with improved training. Kimina-Prover [4] and Goedel-Prover-V2 [5] explore reinforcement learning. These systems benefit from Lean's kernel guarantee: if the proof type-checks, it is correct. Our audit demonstrates why this guarantee matters.

SMT-based approaches. MiniF2F-Dafny [7] (POPL 2026) translates problems to Dafny at 55.7%. Dafny's verification condition generation provides stronger guarantees than raw Z3 but weaker than Lean's kernel. Our findings suggest that any SMT-based approach is vulnerable to encoding failures unless independently audited.

Proof integrity. AlphaProof [8] (Nature 2025) notably chose Lean kernel verification over solver-based approaches for IMO problems. Our audit validates this design choice. FormalMATH [6] extends formalization to broader domains but, as our certificate audit shows, formalization alone does not guarantee proof validity.

LLM evaluation skepticism. The PAJAMA study [11] and "Time to Impeach LLM-as-a-Judge" critique [12] demonstrated systematic over-reporting in LLM evaluation. Work on debate-based verification [13] similarly highlights gaps between perceived and actual quality. Our findings extend this skepticism to formal verification: even in a domain where correctness is supposedly checkable, unchecked results are unreliable.

Knuckledragger. Developed by Philip Zucker [9], Knuckledragger wraps Z3 in a proof-carrying discipline via opaque `Proof` objects. We are, to our knowledge, the first to use it for large-scale LLM theorem proving—and the first to document its limitations when LLMs generate trivial certificates.

7 Discussion

Recommendations. Based on our audits, we make three recommendations for the field:

1. **Always audit.** Report both solve rates and independently audited valid rates. A certificate rate is better than a raw solve rate, but an audited certificate rate is better still.
2. **Report certificate rates AND audit rates.** Certificate rates alone are misleading—our FormalMATH audit shows that 46.7% certificate rate corresponds to 7.4% valid rate. The gap between these numbers is the real trust gap.
3. **Focus on decomposition.** The proof backend is not the bottleneck. Research effort should target LLMs' ability to decompose theorems into non-trivial formal sub-lemmas.

Honest positioning. Table 6 places our results alongside published systems. We are not claiming state-of-the-art performance. We are claiming that honest measurement produces much lower numbers than currently reported for SMT-based approaches.

Table 6: Comparison with published systems on miniF2F (test split). *Lean kernel provides built-in certification; reported rates *are* certificate rates. †Audited valid rate, not raw certificate rate.

System	**Backend**	**Ref.**	**Rate (%)**
Seed-Prover	Lean 4*	[2]	∼100
HILBERT	Lean 4*	[3]	99.2
Kimina-Prover	Lean 4*	[4]	92.2
Goedel-Prover-V2	Lean 4*	[5]	90.4
AlphaProof	Lean 4*	[8]	—
MiniF2F-Dafny	Dafny/Z3	[7]	55.7
Ours (reported cert)	KD/Z3		89.3
Ours (audited valid)†	KD/Z3		∼8–21

What certificates get right. We do not want to dismiss certificates entirely. Knuckledragger certificates eliminate an entire class of failures—vacuous proofs—that constitutes 40% of Z3-only invalidity. This is a genuine and important improvement. The problem is that certificates are necessary but not sufficient: they catch one failure mode while LLMs adapt by producing a different one (trivial certificates).

Limitations. Our manual audits were conducted by a single author. The automated audit uses heuristic checks that may misclassify some proofs. The FormalMATH certificate audit covers 54 problems, which may not represent the full distribution. Knuckledragger itself relies on Z3's proof-producing mode, which is sound but incomplete—some valid proofs may fail to certify. Finally, our findings apply to SMT-based approaches; Lean-based systems with kernel checking are not subject to the same critique.

The decomposition problem is the real frontier. Our audits converge on a single conclusion: the hard problem in LLM theorem proving is not proof verification but theorem decomposition. Until LLMs can reliably break theorems into non-trivial formal sub-lemmas, benchmark numbers—whether based on solver output, certificate rates, or any other automated metric—will overstate actual proving capability. The field needs benchmarks and methods that specifically target decomposition, not just end-to-end solve rates.

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